

COMP 3311

DATABASE MANAGEMENT SYSTEMS

LECTURE 6 EXERCISES

RELATIONAL DATABASE DESIGN: NORMALIZATION

EXERCISE 1

Given: $R(A, B, C, D, E)$ $F = \{A \rightarrow BC\}$

Decomposition: $R_1(A, B, C)$ and $R_2(A, D, E)$

a) Is the decomposition lossless? **Why?** (iff $R_1 \cap R_2 \rightarrow R_1$ or $R_1 \cap R_2 \rightarrow R_2$)

Yes The common attribute A is a key for R_1 .

b) Is the decomposition dependency preserving? **Why?** (iff $(\cup F_i)^+ = F^+$)

Yes $A \rightarrow BC$ is preserved in R_1 .

c) Is the decomposition $R_1(A, B, C)$ and $R_2(C, D, E)$ lossless? **Why?**

No C is not a key for any relation.

EXERCISE 2

Given: $R(A, B, C, D, E)$ $F = \{A \rightarrow BC, CD \rightarrow E, B \rightarrow D, E \rightarrow A\}$

Decomposition: $R_1(A, B, C)$ and $R_2(A, D, E)$

a) Is the decomposition lossless? **Why?** (iff $R_1 \cap R_2 \rightarrow R_1$ or $R_1 \cap R_2 \rightarrow R_2$)

Yes The common attribute A is a key for R_1 .

b) Is the decomposition dependency preserving? **Why?** (iff $(\cup F_i)^+ = F^+$)

No We lose $CD \rightarrow E$ and $B \rightarrow D$.

EXERCISE 3

a) Given: $R(A, B, C, D)$ $F = \{AB \rightarrow CD, B \rightarrow C\}$

Is R in 2NF? **Why?**

Candidate keys: AB $AB^+ = \{A, B, C, D\}$ $B^+ = \{B, C\}$

No For $B \rightarrow C$, B is a proper subset of the key AB and C is non-prime.

So, R is not in 2NF.

b) Given: $R(A, B, C, D)$ $F = \{AB \rightarrow CD, C \rightarrow D\}$

Is R in 2NF? **Why?**

Candidate keys: AB $AB^+ = \{A, B, C, D\}$ $C^+ = \{C, D\}$

Yes For $C \rightarrow D$, C is not a proper subset of the key.

So, R is in 2NF.

2NF

R is in 2NF *if and only if*

For each FD: $X \rightarrow A$ in F^+ :

$A \in X$ (trivial FD) **or**

X is **not** a proper subset of
a candidate key for R **or**

A is a **prime attribute** for R.

EXERCISE 4

Identify the candidate key(s) and the current highest normal form for each of the following relation schemas given their corresponding FDs.

a) $R(A, B, C, D, E)$ $F = \{A \rightarrow B, C \rightarrow D\} = F^+$

$A^+ = \{A, B\}$

$C^+ = \{C, D\}$

Candidate keys: ACE

⇒ For $A \rightarrow B$ and $C \rightarrow D$

- i. A and C are proper subsets of the candidate key ACE (both FDs fail 1st 2NF test).
- ii. both B and D are not prime attributes of R (both FDs fail 2nd 2NF test).

✎ Both FDs violate 2NF.

Normal form: 1NF

2NF

R is in 2NF *if and only if*

For each FD: $X \rightarrow A$ in F^+ :

$A \in X$ (trivial FD) **or**

X is not a proper subset of a candidate key for R **or**

A is a prime attribute for R.

3NF

R is in 3NF *if and only if*

For each FD: $X \rightarrow A$ in F^+ :

$A \in X$ (trivial FD) **or**

X is a superkey for R **or**

A is a prime attribute for R.

EXERCISE 4 (CONT'D)

Identify the candidate key(s) and the current highest normal form for each of the following relation schemas given their corresponding FDs.

b) $R(A, B, C)$ $F = \{AB \rightarrow C, C \rightarrow B\} = F^+$

$AB^+ = \{A, B, C\}$ $C^+ = \{C, B\}$

Candidate keys: AB, AC

⇒ For $AB \rightarrow C$,
C is a prime attribute of R
(FD passes 2nd 2NF and 3NF tests).

⇒ For $C \rightarrow B$,
B is a prime attribute of R
(FD passes 2nd 2NF and 3NF tests).

☞ Both FDs satisfy 3NF.

Normal form: 3NF

2NF

R is in 2NF *if and only if*

For each FD: $X \rightarrow A$ in F^+ :

$A \in X$ (trivial FD) **or**

X is **not** a proper subset of
a candidate key for R **or**

A is a **prime attribute** for R.

3NF

R is in 3NF *if and only if*

For each FD: $X \rightarrow A$ in F^+ :

$A \in X$ (trivial FD) **or**

X is a **superkey** for R **or**

A is a **prime attribute** for R.



EXERCISE 4 (CONT'D)

Identify the candidate key(s) and the current highest normal form for each of the following relation schemas given their corresponding FDs.

c) $R(A, B, C, F)$ $F = \{AB \rightarrow C, C \rightarrow F\} = F^+$

$AB^+ = \{A, B, C, F\}$ $C^+ = \{C, F\}$

Candidate keys: AB

⇒ For $AB \rightarrow C$

- i. AB is not a proper subset of a candidate key (FD passes 1st 2NF test);
- ii. AB is a superkey for R (FD passes 1st 3NF test).

⇒ For $C \rightarrow F$

- i. C is not a proper subset of a candidate key (FD passes 1st 2NF test);
- ii. C is not a superkey of R (FD fails 1st 3NF test);
- iii. F is not a prime attribute (FD fails 2nd 3NF test).

✎ Both FDs satisfy 2NF.

Normal form: 2NF

2NF

R is in 2NF *if and only if*

For each FD: $X \rightarrow A$ in F^+ :

$A \in X$ (trivial FD) **or**

X is **not** a proper subset of a candidate key for R **or**

A is a **prime attribute** for R.

3NF

R is in 3NF *if and only if*

For each FD: $X \rightarrow A$ in F^+ :

$A \in X$ (trivial FD) **or**

X is a **superkey** for R **or**

A is a **prime attribute** for R.



EXERCISE 5

Decompose $R(A, B, C, D, E, F, G)$ into 3NF relations for the FD set $F = \{AB \rightarrow CD, C \rightarrow EF, G \rightarrow A, G \rightarrow F, CE \rightarrow F\}$.

Attribute closures

$$AB^+ = \{A, B, C, D, E, F\} \quad C^+ = \{C, E, F\} \quad G^+ = \{G, A, F\} \quad CE^+ = \{C, E, F\}$$

Candidate keys: ABG?

BG From $G \rightarrow A$ we can infer $BG \rightarrow AB$ using IR2.

$$BG^+ = \{B, G, A, C, D, E, F\} \quad \text{All FDs violate 3NF!}$$

Canonical cover: $F_C = \{AB \rightarrow CD, C \rightarrow EF, G \rightarrow AF\}$

3NF decomposition: $R_1(\underline{A}, \underline{B}, C, D) \quad R_3(\underline{G}, A, F)$
 $R_2(\underline{C}, E, F) \quad R_4(\underline{B}, \underline{G}) \Rightarrow \text{due to the candidate key}$

EXERCISE 6

Decompose $R(A, B, C, D)$ into 3NF and BCNF relations for each of the following FD sets.

a) $F = \{B \rightarrow C, D \rightarrow A\}$

Attribute closures: $B^+ = \{B, C\}$ $D^+ = \{D, A\}$

Candidate keys: BD

Canonical cover: $F_C = \{B \rightarrow C, D \rightarrow A\}$

3NF Decomposition

$R_1(\underline{B}, C)$

$R_2(\underline{D}, A)$

$R_3(\underline{B}, \underline{D}) \Rightarrow$ due to the candidate key

BCNF Decomposition

$R_1(\underline{B}, C)$

$R_2(\underline{D}, A)$

$R_3(\underline{B}, \underline{D})$

EXERCISE 6 (CONT'D)

Decompose $R(A, B, C, D)$ into 3NF and BCNF relations for each of the following FD sets.

b) $F = \{ABC \rightarrow D, D \rightarrow A\}$

Attribute closures: $ABC^+ = \{A, B, C, D\}$ $D^+ = \{D, A\}$

Candidate keys: ABC, BCD Using IR2: $BCD \rightarrow ABC$

Canonical cover: $F_C = \{ABC \rightarrow D, D \rightarrow A\}$

3NF Decomposition

Already in 3NF

$R(\underline{A}, \underline{B}, \underline{C}, D)$

Dependency preserving?

Yes

Is this BCNF decomposition OK?

$R_1(\underline{A}, \underline{B}, \underline{C})$ and $R_2(\underline{D}, A)$ **No** Why?

BCNF Decomposition

Only $D \rightarrow A$ violates BCNF

$R_1(\underline{B}, \underline{C}, \underline{D})$ $R_2(\underline{D}, A)$

Dependency preserving?

No The FD $ABC \rightarrow D$ is lost.

The FD $ABC \rightarrow D$ is lost, and we need to join R_1 and R_2 on A , but A is not a key of R_2 .
 $\Rightarrow$ **Not a lossless decomposition!**



EXERCISE 7

Given: Sale(customer, store, product, price) and the constraints:

A customer buys from only one store.

There is a unique price for each product in a store, but the same product can have a different price in different stores.

a) What are the FDs implied by the above description?

customer $\rightarrow$ store

store, product $\rightarrow$ price

b) What are the candidate keys?

{customer, product}

Since customer, product $\rightarrow$ store, product (IR2)

and customer, product $\rightarrow$ price (IR3)

Also {customer, product}⁺ = {customer, product, store, price}

EXERCISE 7 (CONTD)

3NF

R is in 3NF *if and only if*

For each FD: $X \rightarrow A$ in F^+ :

$A \in X$ (trivial FD) **or**

X is a **superkey** for R **or**

A is a **prime attribute** for R.

Given: Sale(customer, store, product, price)

A customer buys from only one store.

There is a unique price for each product in a store, but the same product can have a different price in different stores.

c) Explain why Sale is not in 3NF.

candidate key: {customer, product}

Both FDs violate 3NF.

$F = \{\text{customer} \rightarrow \text{store}; \text{store, product} \rightarrow \text{price}\}$

The LHSs of the FDs are not superkeys; the RHSs are not prime attributes of Sale.

d) Decompose Sale into 3NF relation schemas.

$R_1(\underline{\text{customer}}, \text{store})$

$R_2(\underline{\text{store, product}}, \text{price})$

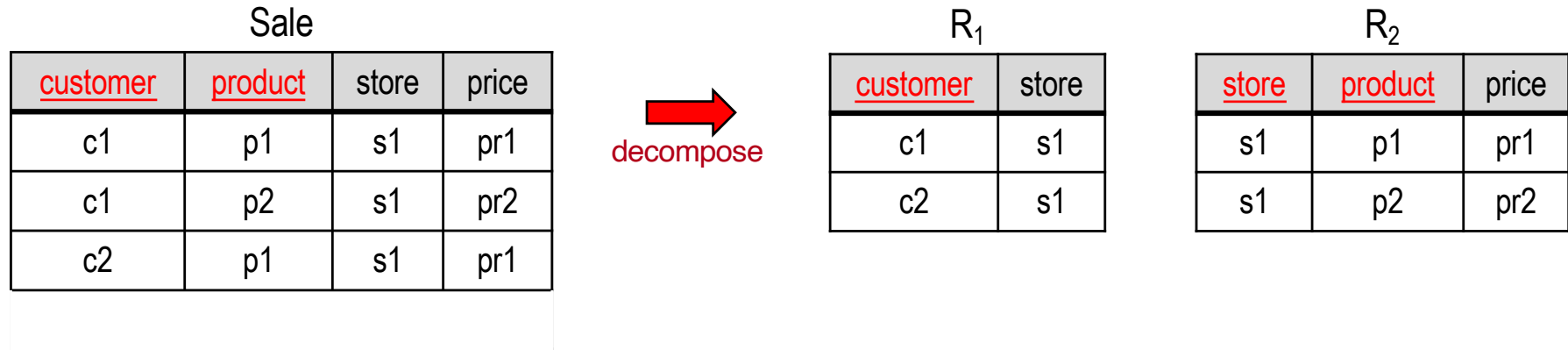
$R_3(\underline{\text{customer, product}})$

e) Is the decomposition dependency preserving? Why?

Yes Each FD is preserved in a relation schema.

EXERCISE 7 (CONTD)

What happens if the decomposition stops at $R_1(\underline{\text{customer}}, \text{store})$ and $R_2(\underline{\text{store}}, \underline{\text{product}}, \text{price})$?
The decomposition is **lossy** since the common attribute **store** is not a key of either relation.



The two decomposed relations do not generate the original one if joined on the common attribute **store**. The **join result contains 4 records instead of 3** as in the original relation.

What is the problem? None of the fragments contains the candidate key (**customer**, **product**).

The solution? Include an additional relation $R_3(\text{customer}, \text{product})$ containing the candidate key in the decomposition.

R_3

<u>customer</u>	<u>product</u>
c1	p1
c1	p2
c2	p1